ECON1050 Tutorial 12

Matrix Algebra III

Week 12 - Semester 2 2026

Xavier Xi Zhang
School of Economics, The University of Queensland
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Today's Plan

- This is our final tutorial
- We do have lecture next week
- Recap: key ideas from Lecture 12
- Tutorial questions: Q1–Q4

Recap: Cramer's Rule

Core idea

For a system

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},$$

Cramer's Rule uses determinants to solve for one variable at a time.

Formula

If $|\boldsymbol{A}| \neq 0$, then

$$x_k = \frac{|\boldsymbol{A}_k|}{|\boldsymbol{A}|},$$

where $\boldsymbol{A}_k$ is formed by replacing the kth column of $\boldsymbol{A}$ with the constants vector $\boldsymbol{b}$.

When useful

Cramer's Rule is useful when we only need one endogenous variable.

Recap: Cramer's Rule for a 3×3 system

System

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Replacement matrices

$$x_1 = \frac{|\boldsymbol{A}_1|}{|\boldsymbol{A}|}, \qquad x_2 = \frac{|\boldsymbol{A}_2|}{|\boldsymbol{A}|}, \qquad x_3 = \frac{|\boldsymbol{A}_3|}{|\boldsymbol{A}|}.$$

Each $\boldsymbol{A}_k$ replaces column k of $\boldsymbol{A}$ with $\boldsymbol{b}$.

$$A_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix}$$

Recap: Minors and cofactors

Minor

For a square matrix $\boldsymbol{A}$, the minor M_{ij} is the determinant of the submatrix obtained by deleting row i and column j.

Cofactor

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

A cofactor exists for every element of a square matrix.

Matrix of cofactors

The matrix of cofactors is denoted by $\boldsymbol{C}^{+}$.

$C_{11} = (-1)^{1+1} \cdot a_{22}$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \Rightarrow C^{+} = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

Recap: The adjoint method

Adjoint / adjugate

The transpose of the matrix of cofactors is called the adjoint:

$$\text{adj}(\boldsymbol{A}) = (\boldsymbol{C}^{+})'.$$

Inverse using the adjoint

If $|\boldsymbol{A}| \neq 0$, then

$$\boldsymbol{A}^{-1} = \frac{1}{|\boldsymbol{A}|}\,\text{adj}(\boldsymbol{A}).$$

Procedure

Find $|\boldsymbol{A}|$, find all cofactors, transpose the cofactor matrix, then divide by $|\boldsymbol{A}|$.

Recap: Solving systems using the inverse

Inverse method

If

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

and $\boldsymbol{A}^{-1}$ exists, then

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}.$$

$Ax = b$

$A^{-1} \cdot Ax = A^{-1} b$

$x = A^{-1} \cdot b$

Non-singularity condition

The inverse exists if and only if

$$|\boldsymbol{A}| \neq 0.$$

Tutorial Q1

Question 1

Solve for x_2 in the following equation system using Cramer's rule.

4 min

$$\begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix} \Leftrightarrow \begin{cases} 2x_1 + 3x_2 - x_3 = -1 \\ 5x_1 + x_2 + x_3 = -2 \\ -x_1 - x_2 + 4x_3 = 8 \end{cases}$$

$$A \quad x \quad = \quad b$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{-49}{-49} = 1$$

$$|A| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = 2 \cdot 1 \cdot 4 + 3 \cdot 1 \cdot (-1) + (-1) \cdot 5 \cdot (-1) - (-1) \cdot 1 \cdot (-1) - 5 \cdot 3 \cdot 4 - 2 \cdot (-1) \cdot 1 = -49$$

$$A_2 = \begin{pmatrix} 2 & -1 & -1 \\ 5 & -2 & 1 \\ -1 & 8 & 4 \end{pmatrix} \Rightarrow |A_2| = (-1)^{1+1} \cdot 2 \cdot \begin{vmatrix} -2 & 1 \\ 8 & 4 \end{vmatrix} + (-1)^{1+2} \cdot (-1) \cdot \begin{vmatrix} 5 & 1 \\ -1 & 4 \end{vmatrix} + (-1)^{1+3} \cdot (-1) \cdot \begin{vmatrix} 5 & -2 \\ -1 & 8 \end{vmatrix}$$

$$= 2 \cdot (-8 - 8) + (20 - 1) - (40 - 2)$$

$$= -49$$

Tutorial Q1: Cramer's Rule

Knowledge used

- Use Cramer's Rule:

$$x_k = \frac{|\boldsymbol{A}_k|}{|\boldsymbol{A}|}.$$

- $\boldsymbol{A}_k$ replaces the kth column of $\boldsymbol{A}$ with $\boldsymbol{b}$.

Solution

$$|\boldsymbol{A}| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = -49.$$

$$|\boldsymbol{A}_1| = 49, \qquad |\boldsymbol{A}_2| = -49, \qquad |\boldsymbol{A}_3| = -98.$$

Therefore,

$$x_1 = \frac{49}{-49} = -1, \qquad x_2 = \frac{-49}{-49} = 1, \qquad x_3 = \frac{-98}{-49} = 2.$$

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Tutorial Q2

Question 2

Consider the following macroeconomic model:

4 mins

Hint: use matrix form

$$
\begin{aligned}
Y &= C + \bar{I} + \bar{G} \\
C &= \alpha + \beta(Y - T) \quad (\alpha > 0,\ 0 < \beta < 1) \\
T &= \gamma + \delta Y \quad (\gamma > 0,\ 0 < \beta < 1)
\end{aligned}
$$

Variables: Y, C, T

Constant: $\bar{I}, \bar{G}, \alpha, \beta, \gamma, \delta$

$$
\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}\begin{pmatrix} Y \\ C \\ T \end{pmatrix} = \begin{pmatrix} & \\ & \\ & \end{pmatrix}
$$

where

Y = national income

C = consumption

$\bar{I}$ = investment, exogenously determined

$\bar{G}$ = government spending, exogenously determined

$$
\begin{aligned}
Y = C + \bar{I} + \bar{G} &\iff Y - C + 0 = \bar{I} + \bar{G} \\
C = \alpha + \beta(Y - T) &\iff -\beta Y + C + \beta T = \alpha \\
T = \gamma + \delta Y &\iff -\delta Y + 0 + T = \gamma
\end{aligned}
$$

$$
\begin{pmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{pmatrix}\begin{pmatrix} Y \\ C \\ T \end{pmatrix} = \begin{pmatrix} \bar{I} + \bar{G} \\ \alpha \\ \gamma \end{pmatrix}
$$

Using Cramer's rule, determine the equilibrium value T^*.

$$\begin{pmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{pmatrix} \begin{pmatrix} Y \\ C \\ T \end{pmatrix} = \begin{pmatrix} \bar{I}+\bar{G} \\ \alpha \\ \gamma \end{pmatrix}$$

$$A \cdot x = b$$

Cramer's rule: $T = \dfrac{|A_3|}{|A|} = \dfrac{\begin{vmatrix} 1 & -1 & \bar{I}+\bar{G} \\ -\beta & 1 & \alpha \\ -\delta & 0 & \gamma \end{vmatrix}}{\begin{vmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{vmatrix}} = \dfrac{\gamma + (-1)\cdot\alpha\cdot(-\delta) + 0 - (-\delta)\cdot 1\cdot(\bar{I}+\bar{G}) - (-1)\cdot(-\beta)\cdot\gamma - 0}{1 + (-1)\cdot\beta\cdot(-\delta) + 0 - 0 - \delta\cdot(-1)\cdot(-\beta) - 0}$$

$$= \dfrac{\gamma + \alpha\delta + \delta(\bar{I}+\bar{G}) - \beta\cdot\gamma}{1 + \beta\cdot\delta - \beta}$$

Tutorial Q2: Matrix form

Knowledge used

- Rearrange equations so that endogenous variables are on the left.
- Write the system as $\boldsymbol{Ax} = \boldsymbol{b}$.

Solution

Rearrange:

$$Y - C = \bar{I} + \bar{G}, \qquad -\beta Y + C + \beta T = \alpha, \qquad -\delta Y + T = \gamma.$$

Hence

$$\begin{bmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{bmatrix} \begin{bmatrix} Y \\ C \\ T \end{bmatrix} = \begin{bmatrix} \bar{I} + \bar{G} \\ \alpha \\ \gamma \end{bmatrix}.$$

Tutorial Q2: Solving for tax revenue

Knowledge used

- Use Cramer's Rule for the variable T.
- Replace the T column of $\boldsymbol{A}$ with the constants vector.

Solution

$$|\boldsymbol{A}| = \begin{vmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{vmatrix} = 1 - \beta(1-\delta).$$

$$|\boldsymbol{A}_T| = \begin{vmatrix} 1 & -1 & \bar{I} + \bar{G} \\ -\beta & 1 & \alpha \\ -\delta & 0 & \gamma \end{vmatrix} = \delta(\alpha + \bar{I} + \bar{G}) + \gamma(1-\beta).$$

Thus

$$T^* = \frac{\delta(\alpha + \bar{I} + \bar{G}) + \gamma(1-\beta)}{1 - \beta(1-\delta)}.$$

Tutorial Q3

Question 3

Find the inverse using the adjoint method: 6 min

$$A^{-1} = \frac{adj(A)}{|A|}$$

(1) $\begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}$, (2) $\begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix}$,

(3) $\begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}$.

(1) $A = \begin{vmatrix} 5 & 2 \\ 1 & -1 \end{vmatrix} \Rightarrow |A| = 5\cdot(-1) - 2\cdot 1 = -7$

$adj(A) = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}^T = \begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{pmatrix} = \begin{pmatrix} (-1)^{1+1}\cdot(-1) & (-1)^{2+1}\cdot 2 \\ (-1)^{1+2}\cdot 1 & (-1)^{2+2}\cdot 5 \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ -1 & 5 \end{pmatrix}$

$A^{-1} = \frac{\begin{pmatrix} -1 & -2 \\ -1 & 5 \end{pmatrix}}{-7} = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{pmatrix}$

(3) $\begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix} = A \Rightarrow |A| = \begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix}$

$= 1\cdot(-2)\cdot(-10) + 0 + (-2)\cdot 4\cdot 2 - 1\cdot(-2)\cdot(-2) - 0 - 1\cdot 1\cdot 2$

$= -2$

$\text{adj}(A) = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}^T = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$

$= \begin{pmatrix} (-1)^{1+1}\cdot\begin{vmatrix} -2 & 1 \\ 2 & -10 \end{vmatrix} & (-1)^{2+1}\cdot\begin{vmatrix} 0 & -2 \\ 2 & -10 \end{vmatrix} & (-1)^{3+1}\cdot\begin{vmatrix} 0 & -2 \\ -2 & 1 \end{vmatrix} \\ (-1)^{1+2}\cdot\begin{vmatrix} 4 & 1 \\ 1 & -10 \end{vmatrix} & (-1)^{2+2}\cdot\begin{vmatrix} 1 & -2 \\ 1 & -10 \end{vmatrix} & (-1)^{3+2}\cdot\begin{vmatrix} 1 & -2 \\ 4 & 1 \end{vmatrix} \\ (-1)^{1+3}\begin{vmatrix} 4 & -2 \\ 1 & 2 \end{vmatrix} & (-1)^{2+3}\cdot\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} & (-1)^{3+3}\cdot\begin{vmatrix} 1 & 0 \\ 4 & -2 \end{vmatrix} \end{pmatrix}$

$= \begin{pmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{pmatrix}$

$\Rightarrow A^{-1} = \frac{\text{adj}(A)}{|A|}$

$= \frac{\begin{pmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{pmatrix}}{-2}$

$= \begin{pmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{pmatrix}$

Tutorial Q3: Adjoint method I

Knowledge used

- Use
$$\boldsymbol{A}^{-1} = \frac{1}{|\boldsymbol{A}|}\,\text{adj}(\boldsymbol{A}).$$
- For 2×2 matrices, compute determinant and cofactors directly.

Solution

For
$$\boldsymbol{A} = \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \qquad |\boldsymbol{A}| = -7.$$
$$\text{adj}(\boldsymbol{A}) = \begin{bmatrix} -1 & -2 \\ -1 & 5 \end{bmatrix}.$$
Hence
$$\boldsymbol{A}^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{bmatrix}.$$

Tutorial Q3: Adjoint method II

Knowledge used

- The same adjoint formula applies to any non-singular square matrix.
- Check $|\boldsymbol{A}| \neq 0$ before finding the inverse.

Solution

For

$$\boldsymbol{B} = \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix}, \qquad |\boldsymbol{B}| = -2.$$

$$\text{adj}(\boldsymbol{B}) = \begin{bmatrix} -10 & 6 \\ -18 & 11 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{B}^{-1} = \begin{bmatrix} 5 & -3 \\ 9 & -\frac{11}{2} \end{bmatrix}.$$

Tutorial Q3: Adjoint method III

Knowledge used

- For a 3×3 matrix, compute all cofactors.
- Then transpose the cofactor matrix to get the adjoint.

Solution

For

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}, \qquad |\boldsymbol{A}| = -2.$$

The adjoint is

$$\text{adj}(\boldsymbol{A}) = \begin{bmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{bmatrix}.$$

Thus

$$\boldsymbol{A}^{-1} = \begin{bmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{bmatrix}.$$

Tutorial Q4

Question 4

Solve the following system of equations by finding the inverse of the coefficient matrix using the adjoint method

$$\begin{cases} 2x_1 + 3x_2 - x_3 = -1, \\ 5x_1 + x_2 + x_3 = -2, \\ -x_1 - x_2 + 4x_3 = 8. \end{cases}$$

5 mins

$$\Leftrightarrow \begin{pmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix}$$

$$A\,x = b$$

Goal : $x = A^{-1} \cdot b$ $\Leftarrow$ find $A^{-1} = \frac{\text{adj}(A)}{|A|}$

$|A| = -49$ we computed it in Q1.

$$\text{adj}(A) = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} = \begin{pmatrix} (-1)^{1+1}\cdot\begin{vmatrix} 1 & 1 \\ -1 & 4 \end{vmatrix} & (-1)^{2+1}\cdot\begin{vmatrix} 3 & -1 \\ -1 & 4 \end{vmatrix} & (-1)^{3+1}\cdot\begin{vmatrix} 3 & -1 \\ 1 & 1 \end{vmatrix} \\ (-1)^{1+2}\cdot\begin{vmatrix} 5 & 1 \\ -1 & 4 \end{vmatrix} & (-1)^{2+2}\cdot\begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} & (-1)^{3+2}\cdot\begin{vmatrix} 2 & -1 \\ 5 & 1 \end{vmatrix} \\ (-1)^{1+3}\cdot\begin{vmatrix} 5 & 1 \\ -1 & -1 \end{vmatrix} & (-1)^{2+3}\cdot\begin{vmatrix} 2 & 3 \\ -1 & -1 \end{vmatrix} & (-1)^{3+3}\cdot\begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -11 & 4 \\ -21 & 7 & -7 \\ -4 & -1 & -13 \end{pmatrix}$$

$$A^{-1} = \frac{\begin{pmatrix} 5 & -11 & 4 \\ -21 & 7 & -7 \\ -4 & -1 & -13 \end{pmatrix}}{-49} = \begin{pmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{pmatrix}$$

$$X = A^{-1}\cdot b = \begin{pmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5}{49} - \frac{22}{49} - \frac{32}{49} \\ -\frac{3}{7} + \frac{2}{7} + \frac{8}{7} \\ -\frac{4}{49} - \frac{2}{49} + \frac{8\cdot 13}{49} \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Tutorial Q4: Inverse method

Knowledge used

- Write the system as $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
- If $\boldsymbol{A}^{-1}$ exists, then $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.

Solution

$$\boldsymbol{A} = \begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} -1 \\ -2 \\ 8 \end{bmatrix}.$$

$$\boldsymbol{A}^{-1} = \begin{bmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{bmatrix}.$$

Hence

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

Therefore,

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Key takeaways

- Cramer's Rule solves variables using ratios of determinants.
- The adjoint method gives $\boldsymbol{A}^{-1} = \frac{1}{|\boldsymbol{A}|}\,\text{adj}(\boldsymbol{A})$ when $|\boldsymbol{A}| \neq 0$.
- Inverse matrices solve systems through $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.